Dynamic NBA Team Ratings and Pregame Win Probabilities

Model Specification, Calibration, and Historical Evaluation

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Abstract

An NBA forecasting system maintains team ratings and estimates pregame win probabilities using a dynamic multifactor rating model. Rating differences, home court, rest, estimated travel, and time-zone changes determine pregame probabilities. Outcomes, schedule strength, lead retention, and box-score performance determine subsequent rating updates. Five validated full seasons contain 6,150 games. Version 0.3 adds partial inter-season rating carryover, a regularized pregame injury correction, and a separately tracked boosted comparison. Across 3,690 games in three retrospective next-season evaluations, the injury model has log loss 0.5983 and winner accuracy 67.07%; the boosted comparison has log loss 0.6000 and accuracy 67.15%. These include opening games. The original version 0.2 evaluation is retained separately. Historical scores are reconstructed, not measured live performance, and previously observed outcomes cannot serve as a fresh independent test of the new structures.

1 The prediction–update cycle

The model answers two questions in order: how likely is a team to win before its game, and how should that result change its rating afterward? Ratings summarize past games. Rest, travel, and home court adjust the next forecast. Once a game ends, the outcome and eight box-score features determine its rating update. The new rating is then available for later forecasts.

The sections below follow that order. Each quantity is defined before it is used in the final update. The eight features $X_1, \ldots, X_8$ are derived in Section 4; they feed a performance factor P, which appears explicitly in the final Elo equation in Section 6. The dictionary in Appendix B records every symbol and its value where one is fixed or fitted.

Team i is the home team and j the away team when defining a matchup. In a team-specific update, the same equations apply to either side using its own base win probability and outcome. The following is a step-by-step construction of an empirical model; its adjustments are modeling choices whose coefficients are fitted, not consequences of a physical law.

1.1 Methodological basis and scope

The probability link follows Bradley and Terry's paired-comparison model [1]. With positive strengths $a_i = 10^{r_i/c}$ and $a_j = 10^{r_j/c}$, neutral equal-context play gives $p_i = a_i/(a_i + a_j)$. The complete equation below is the covariate-adjusted form, with log odds

$$\operatorname{logit}(p_i) = \frac{\ln 10}{c}(r_i - r_j + Hh) - \gamma \ln 10 \ln(F_i/G_i). \tag{1}$$

Thus the Bradley–Terry probability structure remains, while ratings change through a sequential Elo-style residual update; the connection between Elo updates and Bradley–Terry log odds is discussed by Király and Qian [3]. Home court and the context term shift log odds before the game.

Whole-History Rating (WHR), described by Coulom [2], instead jointly estimates time-varying rating trajectories under a Bayesian temporal prior using the available game history. This implementation does not implement that joint maximum-a-posteriori procedure, backward smoothing, or posterior rating distributions. Replaying games to fit shared update constants is not WHR. Its ratings are forward sequential estimates; issued forecasts are never revised using future outcomes. The schedule, lead and box-score multipliers below are empirical project specifications, not derivations from the Bradley–Terry or WHR papers.

2 Step 0: initialize the season

The first available historical season starts all teams at 1500. Version 0.3 retains part of the preceding season's final rating deviation:

$$r_{\text{start}} = 1500 + \rho(r_{\text{previous final}} - 1500), \qquad \rho = 0.5. \tag{2}$$

A final rating of 1600 therefore becomes 1550. Season win records, schedule histories and team performance averages reset. Probability corrections retain earlier player values and team form. The version 0.2 archives retain their original 1500 starts. Carryover supplies an initial strength estimate rather than a preseason roster valuation.

3 Step 1: turn pregame ratings into a win probability

3.1 Start with the rating difference and home court

Let r_i and r_j be the ratings available before tip-off. A basic Elo forecast compares them on a fixed scale c. Home advantage adds Hh to the home-side rating difference, where $h = 1$ at home and $h = 0$ at a neutral venue. With no rest/travel adjustment, the probability would be

$$p_i = \frac{1}{1 + 10^{(r_j - r_i - Hh)/c}}. \tag{3}$$

A higher home rating or a larger home advantage raises the home win probability. The away win probability is always $p_j = 1 - p_i$.

3.2 Add rest and travel context

Write RD for rest in intervening calendar days, TD for venue-city travel in kilometres, and TZ for the time-zone change in hours. Define two positive context indices:

$$F_i = RD_j + e^{|TZ_i|/3} + \nu TD_i, \tag{4}$$

$$G_i = RD_i + e^{|TZ_j|/3} + \nu TD_j. \tag{5}$$

Their ratio compares the two sides' context; it is an empirical adjustment, not a physiological measurement. The fitted coefficient ν determines how much distance contributes. A second coefficient, γ, determines how strongly the log ratio affects the forecast. The complete pregame equation is

$$\boxed{p_i = \frac{1}{1+10^{z_i}}}, \qquad z_i = \frac{r_j - r_i - Hh}{c} + \gamma \ln(F_i/G_i), \qquad p_j = 1 - p_i. \tag{6}$$

Equal context indices give $\ln(F_i/G_i) = 0$, returning the basic Elo forecast. All inputs to this equation are known before the game. Its outcome and box score cannot enter their own prediction.

The base is fixed at 10 and $c = 497.3558141396255$. Fixing the base and rating scale anchors the model's rating units rather than fitting redundant scales. Display rounding does not affect calculations.

3.3 Version 0.3: adjust the published probability for injuries

Write p_i for the base Elo probability and q_i for the default published probability. Define $\sigma(a) = (1+e^{-a})^{-1}$ and $\operatorname{logit}(p) = \ln(p/(1-p))$. The correction is an L2-penalized logistic model with the base log odds as an offset; penalized logistic modeling is discussed in [5, Sections 4.4 and 5.5]. The implemented equation is

$$\boxed{q_i = \sigma\Big(\operatorname{logit}(p_i) + b + \mathbf{w}^{\mathsf{T}}\mathbf{d}_i\Big)}, \qquad q_j = 1 - q_i. \tag{7}$$

The vector $\mathbf{d}_i$ contains standardized pregame covariates: base log odds, status-weighted prior player minutes and value, and explicit coverage indicators. It is separate from the postgame features $X_1, \ldots, X_8$ below. Means, standard deviations, intercept and weights use earlier seasons only and are frozen with a feature-order schema.

Official Out, Doubtful, Questionable and Probable designations are separate [12]. They are categorical predictors, not fixed probabilities of playing or clinician estimates. For each player, estimate expected minutes $\bar{m}$ with a normalized exponential mean over earlier played observations, half-life ten observations. The descriptive player score uses the Game Score formula documented by Sports Reference [4]:

$$\begin{aligned}\text{PlayerScore} = {} & PTS + 0.4FGM - 0.7FGA - 0.4(FTA - FTM) \\ & + 0.7OREB + 0.3DREB + STL + 0.7AST + 0.7BLK - 0.4PF - TOV.\end{aligned} \tag{8}$$

BLK and PF are blocks and personal fouls. With discounted prior sums V of PlayerScore and M of minutes, the shrunk value per minute is

$$v_p = \frac{V + 80(0.3)}{M + 80}. \tag{9}$$

For team t and status s, define the two contributions

$$A_{t,s} = \sum_{a \in (t,s)} \frac{\bar{m}_a}{240}, \qquad B_{t,s} = \sum_{a \in (t,s)} \frac{\bar{m}_a v_a}{80}. \tag{10}$$

The eight status features are $A_{i,s} - A_{j,s}$ and $B_{i,s} - B_{j,s}$ for the four statuses. Ten further coverage features comprise differences in submitted flags, unmapped counts, missing-history counts and

valued-player counts, plus each side's submitted flag, unmapped count and missing-history count separately. Together with base log odds, the correction has 19 inputs. For raw input u_k, the fitted standardization is $d_k = (u_k - \mu_k)/s_k$, where μ_k, s_k use training rows only and scales below 10^{-10} are replaced by one. The base log odds appear both as a fixed offset and as a standardized covariate, allowing fitted recalibration. The constants 240 and 80 set feature scales; 80 minutes at value 0.3 supplies a player-value prior. Coverage features distinguish submitted, unsubmitted, unmapped and no-prior-history cases. Missing records never imply observed health or zero playing time. These descriptive player values are not RAPM.

If no verified report covers the game, use $q_i = p_i$. Partially submitted reports use explicit unknown indicators. Live collection rejects future-dated or more-than-16-hour-old reports. The Elo update always uses p_i, not q_i: the injury correction does not feed back into ratings.

3.3.1 From source reports to issued forecasts

The scheduled collector first reads the official report index, or checks at most four distinct already-due public quarter-hour PDF versions if the index is unavailable. It parses the PDF table and actual header publication time, rejects future or more-than-16-hour-old documents, and matches game date, teams and matchup. Normalized player names map only when unambiguous to ESPN athlete identifiers. Earlier completed ESPN player box scores supply minutes and PlayerScore; invalid team/player totals are excluded from player history rather than interpreted as zero minutes [9, 8]. Expected-minute smoothing, shrinkage constants and feature scales are project choices, not independently established player-impact estimates or injury severity measurements.

Historical forecasts freeze the whole Eastern-calendar slate at the earlier of 10 a.m. and one hour before its first tip-off; no player performance from that slate enters its own morning forecasts. Live revisions use information collected at their actual issuance time, never a backdated timestamp. Every issued revision preserves the report URL, publication time, hash, coverage, model fingerprint and numerical input vector. Partial submissions use unknown indicators; absent verified matchup coverage gives $q_i = p_i$. The two-hour collection cadence can lag later NBA report changes, which is why the dashboard exposes the report timestamp. Game results update the Elo state through the base residual $y_i - p_i$; the published injury probability q_i is used for forecasts and their evaluation, not that rating residual.

3.4 A separately tracked boosted comparison

Shallow histogram gradient-boosted trees produce $q_{B,i}$ in an internal research comparator. Gradient boosting follows the additive stagewise framework of Friedman [7]; the histogram implementation is Scikit-learn [13]. Inputs include base log odds, injury features, earlier form, schedule and rotation summaries. For each previous team game, store net efficiency, signed margin minus $7.5\,\mathrm{logit}(p)$, the eight raw box-score features, points for and against, and estimated possessions. Use normalized exponential means with half-lives 5, 10 and 20 for the first two quantities and half-life 10 for the remainder. These form summaries persist across seasons.

Schedule inputs are clipped rest (maximum seven days), back-to-back indicators, previous games in four calendar days, distance in thousands of kilometres, signed and absolute time-zone movement, and each side's rest. Rotation inputs compare the last two observed minute-share distributions using $\sum_a \min(s_{a,1}, s_{a,2})$, measure the last rotation's minutes belonging to observed prior-season

players, and measure the cumulative-minute fraction of the three highest-minute players missing from the last rotation. History-coverage flags accompany these quantities. This is delayed rotation information, not a current-game lineup or preseason roster oracle.

For numerical feature vector $\mathbf{f}$, the comparison is

$$q_{B,i} = (1-\omega)p_i + \omega\,\sigma\left(a_0 + \sum_{t=1}^{T} h_t(\mathbf{f}_i)\right). \tag{11}$$

Each h_t is a trained tree with its learning rate included in leaf values. Earlier validation selects the settings and blend weight ω. Both corrections share the same Elo rankings; boosting is a challenger, not the public dashboard forecast.

4 Step 2: derive eight features from the completed box score

Only after the game finishes do its box-score counts become inputs to the rating update. Let O denote the opponent. First estimate possessions:

$$\text{Poss} = FGA + 0.44FTA - OREB + TOV. \tag{12}$$

FGA and FTA are field-goal and free-throw attempts, $OREB$ is offensive rebounds, and TOV is summed player turnovers. The coefficient 0.44 approximates the possession cost of free throws; offensive rebounds are subtracted because they extend a possession. Opponent possessions use the same expression with opponent counts.

The eight features describe scoring efficiency, shooting, rebounding, ball movement, ball security, and steals:

$$X_1 = \frac{100(PTS - PTS_O)}{\text{Poss}} \qquad \text{net efficiency,} \tag{13}$$

$$X_2 = \frac{FGM + 0.5(3PM)}{FGA} \qquad \text{effective field-goal percentage,} \tag{14}$$

$$X_3 = \frac{PTS}{2(FGA + 0.44FTA)} \qquad \text{true shooting percentage,} \tag{15}$$

$$X_4 = \frac{OREB}{OREB + DREB_O} \qquad \text{offensive-rebound rate,} \tag{16}$$

$$X_5 = \frac{DREB}{DREB + OREB_O} \qquad \text{defensive-rebound rate,} \tag{17}$$

$$X_6 = \frac{AST}{\max(TOV, 1)} \qquad \text{assist-to-turnover ratio,} \tag{18}$$

$$X_7 = \frac{TOV}{FGA + 0.44FTA + TOV} \qquad \text{turnover fraction,} \tag{19}$$

$$X_8 = \frac{STL}{\text{Poss}_O} \qquad \text{steal rate.} \tag{20}$$

Here PTS means points, FGM means field goals made, $3PM$ means three-point makes, $DREB$ means defensive rebounds, AST means assists, and STL means steals. Rates are fractions rather than percentages on a 0–100 scale. X_1 uses a shared-possession net-efficiency approximation. The two rebound rates pair each team category with the opposite opponent category. A zero-turnover game uses one in the assist-ratio denominator; other nonpositive denominators fail validation.

5 Step 3: combine the features into a performance factor

5.1 Scale and weight the eight features

The raw features have different units and ranges. Denote their fixed scaled versions by lowercase x:

$$(x_1, \ldots, x_8) = (X_1/10, X_2, X_3, X_4, X_5, X_6/3, -X_7, X_8). \tag{21}$$

Dividing net efficiency by 10 and assist ratio by 3 sets their relative scale. The negative sign on turnover fraction makes fewer turnovers contribute a larger score. These scales were fixed before fitting.

The fitted weights $\alpha_1, \ldots, \alpha_8$ are positive and sum to one. The weighted box-score total is

$$\text{Score} = \sum_{k=1}^{8} \alpha_k x_k. \tag{22}$$

Thus every X_k affects the total through its scaled value x_k and its weight α_k. A large weight is not itself a causal claim or a scale-free measure of feature importance.

5.2 Compare this game with the team's previous games

Convert the weighted total into a bounded game-performance score:

$$\text{Perf} = \text{clip}\left(\frac{1}{1 + e^{-\text{Score}}}, 10^{-6}, 1 - 10^{-6}\right). \tag{23}$$

The sigmoid avoids singularities and accepts negative efficiency values. Clipping limits its result to the stated lower and upper bounds. For a team with n earlier games, its reference performance is

$$\text{PriorPerf} = \frac{1}{n} \sum_{g=1}^{n} \text{Perf}_g. \tag{24}$$

This average excludes the current game. The relative-performance factor is

$$\boxed{P = \text{clip}\left(\frac{\text{Perf}}{\text{PriorPerf}}, 0.25, 4\right)}. \tag{25}$$

$P > 1$ means better box-score performance than the team's prior average; $P < 1$ means worse. Season openers use $P = 1$, because no previous team games are available. Stored data calls Perf the aggregate game performance factor (AGPF) and P the performance adjustment factor (PAF).

The complete feature path is therefore

$$X_1, \ldots, X_8 \;\longrightarrow\; x_1, \ldots, x_8 \;\longrightarrow\; \text{Score} \;\longrightarrow\; \text{Perf} \;\longrightarrow\; P. \tag{26}$$

This factor is now ready to enter the Elo update; it is not an additional input to the already-issued probability for the same game.

6 Step 4: assemble the final Elo update

6.1 Start with the outcome relative to the prediction

For the team being updated, let $y = 1$ for a win and $y = 0$ for a loss, and let p be its pregame probability from Equation (6). The basic Elo update is

$$r_{\text{new}} = r + K(y - p). \tag{27}$$

K controls update size. An unexpected win produces a larger positive error than an expected win; a loss gives a negative error. The model multiplies this base update by three adjustments. Define the result sign $s = 2y - 1$, equal to $+1$ for a win and -1 for a loss.

6.2 Adjustment 1: strength of schedule

A team's prior smoothed win rate is

$$W_t = \frac{w_t + 1}{n_t + 2}, \tag{28}$$

where w_t and n_t count only wins and games before the forecast cutoff. Let $\overline{W}_{\text{opp history}}$ be the mean smoothed win rate of its opponent's previous opponents, with repeated opponents counted in proportion to their frequency. Then

$$S = \frac{2W_{\text{opp}} + \overline{W}_{\text{opp history}}}{3}, \qquad M_S = 1 + s\beta S. \tag{29}$$

Empty histories use 0.5. The fitted coefficient β controls this schedule multiplier. Its value is calculated from prior results, not the current game's box score.

6.3 Adjustment 2: lead retention

Let LL and LL_O be the two teams' largest leads, and PTS and PTS_O their final scores. Define

$$L = \frac{|PTS - PTS_O|}{\max(LL, LL_O)}, \qquad M_L = 1 + \phi \log_2(1 + L). \tag{30}$$

For valid non-tied games, $0 < L \leq 1$. This compares the final margin with the largest advantage held by either side. It remains defined even when one team never led. Peak leads below the final margin fail validation. The fitted coefficient ϕ sets the influence of this adjustment.

6.4 Adjustment 3: box-score performance

Use the factor P already derived from the eight features:

$$M_P = 1 + s\lambda(P - 1). \tag{31}$$

For a winner, $P > 1$ increases the positive update and $P < 1$ reduces it. For a loser, $P > 1$ reduces the magnitude of its negative update and $P < 1$ increases it. With P clipped to $[0.25, 4]$ and $0 \leq \lambda \leq 0.25$, the multiplier cannot become negative. This is exactly where the X features enter Elo.

6.5 The complete equation

Multiplying the three adjustments gives the final model:

$$\boxed{r_{\text{new}} = r + K(y - p)\, M_S\, M_L\, M_P}. \tag{32}$$

Substituting their definitions makes every contribution explicit:

$$\boxed{r_{\text{new}} = r + K(y - p)(1 + s\beta S)(1 + \phi \log_2(1 + L))(1 + s\lambda(P - 1))}. \tag{33}$$

The probability p uses prior information; L and P use the just-completed game. The dependence of the final multiplier on the box score is

$$(X_1, \ldots, X_8) \longrightarrow P \longrightarrow M_P \longrightarrow r_{\text{new}} \longrightarrow \text{later-game probabilities}. \tag{34}$$

Both teams' updates use their pregame ratings and are computed before either is committed. Asymmetric updates can change the league mean. After the historical slate, subtracting that mean minus 1500 from every rating recenters the league without changing pairwise rating differences. Live state recenters after individual completed games. Inter-season initialization follows Equation (2); it does not change the within-season equation.

6.6 Worked example from the retained version 0.2 replay

Consider Detroit Pistons against Washington Wizards, game `401810054` in the 2025–26 archive. This is an existing reconstructed forecast in the original `snapshot.json`, retained beside the new version 0.3 archive, not an issued live prediction. For the home team, the stored eight box-score features and frozen weights give

$$\begin{aligned} \text{Score} &= 0.611115, \\ \text{Perf} &= 0.648195, \\ \text{PriorPerf} &= 0.656950, \\ P &= 0.648195/0.656950 \approx 0.986673. \end{aligned}$$

The team won ($y = 1$), despite performing slightly below its prior box-score average ($P < 1$). Its forecast was $p = 0.682259$, schedule strength $S = 0.285256$, and lead retention $L = 0.153846$. The resulting multipliers are

$$M_S = 1.279137, \qquad M_L = 1.412902, \qquad M_P = 0.996668.$$

Using the full-precision inputs,

$$\Delta r = K(1 - p)M_S M_L M_P = 6.992450, \qquad r_{\text{new}} = 1571.901646 + 6.992450 = 1578.894096.$$

This last rating is before league recentering; the stored post-slate rating also includes the common recentering shift. The example shows how a win can still receive a slightly smaller performance multiplier when its box-score performance is below the team's own baseline.

7 Data construction

Free ESPN-derived schedule, team-box-score, and play-by-play parquet files from SportsDataverse provide the initial history [8]. Primary ESPN game summaries replace incomplete or inconsistent

archival records [9]. GeoNames supplies venue-city coordinates and IANA time zones [10]. Each of the five seasons contains 1,230 completed standard regular-season games, 30 teams, and 82 games per team. Preseason, playoffs, All-Star games, and the NBA Cup final are excluded; Cup games that count toward the regular season remain included.

Validation requires unique game/team identities, matching opponent rows and header scores, finite nonnegative integer counts, made shots no greater than attempts, scoring identities, plausible assists and steals, valid feature denominators, and peak leads consistent with final margins. Missing values are not converted to zero. Largest leads can be recovered from a score path only when its final scores match the header. Inconsistent team aggregates can be replaced by complete player rows with matching points and shooting identities. Turnovers consistently mean summed player turnovers, excluding provider non-player team turnovers.

Season	Accepted games	Primary replacements	Unresolved failures
2021–22	1230	132	0
2022–23	1230	158	0
2023–24	1230	201	0
2024–25	1230	165	0
2025–26	1230	217	0

Table 1: Source audit. The 873 primary replacements mostly address incomplete archived play-by-play; they are not a count of corrupt final box scores.

All required numerical formula inputs are complete. Repair actions, availability proxies, source URLs, retrieval times where available, and SHA-256 hashes are stored with the data or collection manifests. This establishes internal consistency, not perfect measurement of every real-world quantity. Travel is estimated between successive venue cities, not actual itineraries. Rest counts intervening Eastern dates, so back-to-backs have zero. Openers receive three rest days and zero travel/time-zone change, with an initialization flag. Both venue offsets are evaluated at the current tip-off instant, preventing a daylight-saving change alone from creating geographic travel. Ambiguous or unknown cities cause validation failure.

7.1 Chronology and forecast availability

A prediction uses only information available at its recorded forecast cutoff. The home-win outcome is observed afterward. The historical replay freezes all forecasts for an Eastern-calendar daily slate at the earlier of 10:00 America/New_York and one hour before its first tip-off. It requires all prior slates to have completed before that cutoff; current-slate results cannot enter its forecasts. Ratings update after the slate. A team cannot appear twice on one slate, and duplicate games are rejected.

Completion availability is proxied by terminal EndGame wallclock timestamps, rather than later edited play timestamps. Seventeen games use dated final-recap publication times. One game with source play dates shifted by a day uses an explicitly flagged conservative tip-off-plus-eight-hours availability bound; this is not an observed completion time. Replays check these proxies against the next cutoff but cannot establish when historical corrections first reached a live feed. Genuinely issued forecasts record their actual issue time, data cutoff, and model version in a separately preserved ledger.

8 Reference version 0.2 parameter estimation

Training uses 2021–22 through 2023–24; 2024–25 is validation and 2025–26 is a held-out test. The objective is mean binary log loss,

$$\mathcal{L}(\theta) = -\frac{1}{N}\sum_{g}[y_g \ln p_g(\theta) + (1 - y_g)\ln(1 - p_g(\theta))]. \tag{35}$$

A game is scored after both teams have played ten prior games, giving 3,220 training observations. Earlier games still initialize histories, update ratings, and retain forecasts. Every candidate recomputes parameter-dependent ratings and game-performance history. An accelerated Numba replay is checked against the canonical Python engine before fitting and at the fitted solution, to probability tolerance 10^{-11}.

SciPy differential evolution with L-BFGS-B polishing [11] fits 14 free dimensions: seven scalar coefficients and seven softmax logits. With the eighth logit fixed to zero, $\alpha_k = \exp(u_k)/\sum_l \exp(u_l)$ and $u_8 = 0$. Travel is optimized on a base-10 logarithmic scale. Settings are random seed 42, population multiplier 8, maximum 150 generations, relative tolerance 10^{-5}, one worker, and polishing enabled. The run converged in 44 generations and 5,520 objective evaluations. Training log loss decreased from 0.637588 at the reference starting configuration to 0.628210. That release's constants remain frozen; new structures are versioned and evaluated separately below.

Parameter	Search interval	Fitted value
K	$[1, 80]$	12.217347
H	$[0, 180]$	41.552847
γ	$[0, 0.5]$	0.164724
ν	$10^{[-8,-1]}$	0.000109735
ϕ	$[0, 2]$	2.000000
λ	$[0, 0.25]$	0.250000
β	$[0, 1]$	0.978548
$u_1, \ldots, u_7$	$[-5, 5]$	softmax weights below

Table 2: Reference version 0.2 constants. Full precision is retained in the versioned configuration. Base 10, c, initial rating 1500, and time-zone scale 3 are fixed.

α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8
0.161954	0.004365	0.006380	0.002774	0.686992	0.073041	0.007926	0.056568

The lead and performance coefficients reached their upper bounds. This is a converged bounded solution, not proof of a global optimum or uniquely identified causal effects. Fixed feature scales also influence weight interpretation. Future constraint changes require a new model version and new prospective evidence; the observed test season is no longer untouched for such revisions.

9 Reference version 0.2 historical evaluation

Probability quality is assessed by log loss, Brier score $N^{-1}\sum_g (p_g - y_g)^2$, and ten equal-width calibration bins with counts, mean forecasts, and observed home-win fractions. Lower log loss

and Brier score are better. These are strictly proper probability scoring rules [6]; winner accuracy discards confidence by applying a threshold. Winner accuracy is secondary, choosing home at exactly $p = 0.5$. It is not an accuracy percentage for probabilities.

Season	Role	Scored	Log loss	Brier	Winner accuracy
2021–22	Training	1073	0.6298	0.2191	66.64%
2022–23	Training	1073	0.6457	0.2273	62.26%
2023–24	Training	1074	0.6092	0.2109	66.01%
2024–25	Validation	1075	0.6043	0.2088	66.42%
2025–26	Held-out test	1075	0.5925	0.2035	68.56%

Table 3: Frozen-model chronological replay. All 1,230 forecasts per season are retained; primary scoring excludes games before both teams' ten-game warmup.

Across five seasons, 5,370 games are scored and 780 are warmup-only. Test ratings continue updating after observed outcomes, matching an adaptive rating system while coefficients remain frozen. Training scores are in-sample. A single held-out season does not establish performance across all future seasons, and sparse extreme-probability bins limit calibration inference. The recorded bin summaries are descriptive; no uncertainty-adjusted calibration or repeated-season confidence claim is made.

10 Simple Elo comparison and chronological diagnostics

A comparator estimates only K and H, with the same fixed c, season initialization, cutoff, and warmup as the full model. Its equations are

$$p_i = [1 + 10^{(r_j - r_i - Hh)/c}]^{-1}, \qquad r_{i,\text{new}} = r_i + K(y_i - p_i), \quad r_{j,\text{new}} = r_j - K(y_i - p_i). \tag{36}$$

Fatigue, schedule, lead, and performance coefficients are zero. Both models fit training log loss with identical bounds on K and H, seed 42, and the same differential-evolution and polishing settings. Comparator predictions are checked against the canonical engine. Data and scored games are paired. This diagnostic was added after viewing original results; it is not a new untouched test or prospective evaluation.

First, compare the original three-season frozen configuration with simple Elo fitted on those same seasons:

Season	Model	Scored	Log loss	Brier	Accuracy
2024–25	Multifactor	1075	0.6043	0.2088	66.42%
2024–25	Simple Elo	1075	0.6137	0.2129	65.95%
2025–26	Multifactor	1075	0.5925	0.2035	68.56%
2025–26	Simple Elo	1075	0.6018	0.2073	66.79%

Second, expanding-window checks fit the first available season and evaluate the next, then add one training season per fold. Only the training prefix enters either objective. Full-model equations and bounds remain fixed; the three-season full fit is reused from the original release. Other fits are research configurations and do not overwrite archived release predictions.

Evaluation	Train seasons	Scored	Full LL	Simple LL	Full BS	Simple BS
2022–23	1	1073	0.6469	0.6522	0.2280	0.2300
2023–24	2	1074	0.6105	0.6168	0.2114	0.2145
2024–25	3	1075	0.6043	0.6137	0.2088	0.2129
2025–26	4	1075	0.5920	0.6013	0.2033	0.2071

All added optimizations converged. The full model has lower log loss and Brier score in all four folds. Winner accuracy improves in three folds but falls from 62.63% to 61.32% in 2022–23. Probability-score improvement therefore does not imply improvement in every classification statistic.

Paired exploratory uncertainty estimates use 10,000 bootstrap resamples of whole Eastern-calendar daily slates. Each sampled slate retains its eligible games, and differences are weighted by the total sampled game count. This preserves within-slate dependence but does not model dependence across days or fitting uncertainty; intervals have no multiple-comparison adjustment. For the original frozen model on 2025–26, the log-loss difference (full minus simple) is -0.00928, with exploratory 95% interval $[-0.01417, -0.00395]$. The analogous expanding-window interval crosses zero in 2022–23 and is below zero in the other three folds. These intervals are descriptive evidence rather than a guarantee of future improvement.

Across the four full fits, K ranges from 11.134 to 13.085, H from 33.368 to 47.570, and γ from 0.1647 to 0.1829. Individual travel and performance weights are less stable: ν ranges from 1.32×10^{-7} to 1.10×10^{-4}, and the defensive-rebound weight α_5 ranges from 0.4795 to 0.9610. Lead and performance coefficients remain at their upper bounds in every fit. The checks support a modest repeated probability-score advantage for the model family, while leaving uncertainty about unique feature effects and overfitting. Attributing improvement to individual features would require an ablation study.

The original v0.2 configuration and issued ledger are retained as reference records. Source hashes, per-fold fit reports, metrics, paired predictions, and intervals are saved under the comparison release directory. Future issued forecasts will provide prospective monitoring. The historical outcomes were already available during development, so these expanding-window checks must not be presented as newly untouched evidence.

11 Version 0.3 fitting and integrated evaluation

The release core coefficients use all five complete earlier seasons, with the original bounded objective and ten-game warmup. Carryover is fixed at 0.5. Probability heads use historically generated base forecasts from each season's earlier-prefix fit, with fixed defaults for the first anchor season. For each outer year, the last earlier season selects settings, earlier seasons fit candidates, and all earlier seasons refit the selected head before the outer year is evaluated.

The injury head minimizes mean binary log loss plus $\eta\|\mathbf{w}_*\|_2^2/2$, including the intercept in $\mathbf{w}_*$. Select $\eta \in \{10^{-4}, 10^{-3}, 10^{-2}, 0.1, 1, 10\}$, with an unchanged-base option. Standardization uses fit rows only. Boosting selects earlier validation winner accuracy, breaking ties by log loss. Candidate depths are 2 or 3, learning rates 0.03 or 0.1, iterations 100 or 250, and L2 penalties 1 or 10; minimum leaf size is 50, maximum leaves 15, seed 42, with no random early-stopping split. Blend weights are $\{0, 0.25, 0.5, 0.75, 1\}$. The future-season release selects 100 depth-two trees, rate 0.03, L2 penalty 10 and $\omega = 1$; its injury penalty is 10^{-4}.

Across 816 historical game dates, 815 official morning PDFs were retrieved and validated. Player totals/minutes pass checks for 12,235 of 12,300 team-games; invalid player histories are excluded while valid team records remain usable. Reports are matched by date, team and matchup, and their header timestamps must precede the cutoff. All current-slate player updates are withheld until every morning forecast has been captured. This production check corrected five earlier exploratory feature rows affected by overlapping trade/report assignments. Refitting changes some boosted selections; the following results supersede earlier exploratory boosted-accuracy numbers.

Integrated procedure	Games	Log loss	Brier	Accuracy
Carryover Elo	3690	0.603496	0.208434	66.64%
Elo + injuries (default)	3690	0.598309	0.206217	67.07%
Boosted comparison	3690	0.600005	0.206871	67.15%

The public dashboard exposes only Elo + injuries, which has the lowest pooled log loss and Brier score among the tested procedures. The boosted comparator predicts three more winners out of 3,690 games, but has worse probability scores. This is an operational choice for probability quality, not proof that one procedure dominates on all future seasons. The model comparison records remain available for reproducibility.

These three-season retrospective comparisons include opening games and the deployed missing-report fallback. They are not directly comparable with the version 0.2 single-season warmup-filtered 68.56% headline. Outcomes were already studied during development; searching multiple families adds selection uncertainty. Boosting is sensitive to small preparation/refitting changes, reinforcing its role as a challenger. Frozen prospective forecasts provide the next evidence.

The current core fit uses 5,370 eligible games and converges in 37 generations at training loss 0.615970. Rounded constants are $K = 13.748881$, $H = 40.034399$, $\gamma = 0.179560$, $\nu = 0.000095344$, $\beta = 0.708190$, $\phi = 2$ and $\lambda = 0.25$. Full coefficients, scalers, numerical tree nodes, provenance and per-fold reports are stored in versioned JSON. Boundary estimates and changing feature weights remain empirical limitations.

12 Reproducibility and live use

The repository records source URLs/hashes, validation actions, canonical feature tables, frozen constants, optimizer settings, reconstructed forecasts, and comparator predictions. Pregame and postgame columns are explicitly separated. Raw archives and cached responses remain outside Git. Reproduction scripts and operating instructions are documented in the repository. Provider archives can change; manifests identify the source snapshots used for this study. Accelerated and canonical calculations are checked together; cross-platform replay comparisons permit relative/absolute differences of 10^{-12} while preserving exact game identifiers and record structure.

The public Streamlit application displays team ratings, game probabilities, outcomes, evaluation, and historical seasons. GitHub Actions collects the feed every two hours; open pages reread saved snapshots every minute. No fitting or source scraping occurs during page interactions. The collector validates final games, uses a single-writer lock, and atomically publishes processed snapshots. Failure preserves valid prior data and records an update status. Hosting and collection use free services within applicable allowances; provider availability and scheduler timing are not guaranteed.

Live rating state can be reconstructed after source corrections, while its issued-forecast ledger remains immutable. A forecast receives its actual issue time after collection and can be issued only before tip-off. Changed inputs append a revision; identical inputs do not. New-version rows retain the base, injury and boosted probabilities, input vectors, report timestamp/hash and frozen model specification. Old rows are neither overwritten nor given fabricated comparison probabilities. Only the latest matching revision issued before the recorded start is scored. A backfilled completed game with no issued forecast can update ratings but never becomes a live prediction. Primary scoring requires ten observed prior games for each team at issuance. No future outcomes are invented when projecting schedule context.

Historical evaluation freezes daily slates; live collection processes individual completions. Their cutoff policies differ, so the reported historical scores are not a measured accuracy of the live service. Future performance must be assessed from genuinely issued forecasts. Application and source-collection implementation details are not additional predictive evidence.

A Travel and time-zone derivation

For venue latitude φ and longitude ℓ in radians,

$$a_v = \sin^2((\varphi_2 - \varphi_1)/2) + \cos\varphi_1 \cos\varphi_2 \sin^2((\ell_2 - \ell_1)/2), \tag{37}$$

$$TD = 2R \arcsin\sqrt{\mathrm{clip}(a_v, 0, 1)}, \qquad R = 6371 \text{ km}. \tag{38}$$

Neutral-game venue cities are used rather than substituting a franchise home city. If $o(v, t)$ is a venue's dated UTC offset in hours, then $TZ = o(v_{\text{current}}, t_g) - o(v_{\text{previous}}, t_g)$. Exact arenas and actual itineraries are not measured by these proxies.

B Symbol dictionary

The dictionary lists model constants and the quantities used in the construction. Descriptive names replace opaque intermediate symbols: Score is the weighted feature sum, Perf is a bounded game score, PriorPerf is its prior average, and P is their clipped ratio. Constants are rounded for display; the frozen release configuration retains full precision. Dictionary values refer to the current version 0.3 core fit unless a reference version is stated explicitly. The version 0.2 artifacts and forecasts remain unchanged.

Fitted constants and fixed scales

Symbol	Meaning	Value or range
K	The base update coefficient controls how much a game can change Elo.	13.748881
H	Home advantage adds this many Elo points to the home-side rating difference.	40.03439907
c	The fixed rating scale converts Elo differences into probability differences.	497.3558141 (fixed)
γ	The fatigue coefficient weights the logarithm of the rest/travel index ratio.	0.179559595
ν	The travel coefficient weights venue-city distance inside the fatigue indices.	9.534356884e-05
β	The schedule coefficient weights the prior strength-of-schedule adjustment.	0.7081901219
ϕ	The lead coefficient weights the logarithmic lead-retention adjustment.	2
λ	The performance coefficient weights relative box-score performance.	0.25
10	The probability exponent uses a fixed base of ten.	10 (fixed)
3	The fatigue indices divide absolute time-zone change by this fixed hourly scale.	3 hours (fixed)
R	The Haversine calculation uses this fixed Earth-radius approximation.	6371 km

Learned feature weights

Symbol	Meaning	Value or range
α_1	The normalized weight multiplies scaled net efficiency.	0.0006686328527
α_2	The normalized weight multiplies scaled effective field-goal percentage.	0.0008033843254

Symbol	Meaning	Value or range
α_3	The normalized weight multiplies scaled true shooting percentage.	0.008884867219
α_4	The normalized weight multiplies scaled offensive-rebound rate.	0.0007283531314
α_5	The normalized weight multiplies scaled defensive-rebound rate.	0.1613285001
α_6	The normalized weight multiplies scaled assist-to-turnover ratio.	0.8047679813
α_7	The normalized weight multiplies scaled turnover fraction.	0.01518263624
α_8	The normalized weight multiplies scaled steal rate.	0.007635644786

Matchup and rating-update quantities

Symbol	Meaning	Value or range
i, j, O	In the probability equation, i is the home team and j the away team; O denotes an opponent in feature derivations.	Team identifiers
g, t	The index g labels a game; t labels a team in win-rate notation and a time instant in the offset function.	Context dependent
r_i, r_j	These are the team ratings available before the game.	Carryover; first anchor 1500
r_{new}	This is a team rating after its game update, before league recentering.	Varies by game
p_i, p_j	These are the complementary pregame probabilities of a home and away win.	Between 0 and 1; sum 1
z_i	The probability exponent combines the rating difference, home advantage, and fatigue adjustment.	Varies by matchup
h	The venue indicator enables home advantage at a home venue and removes it at a neutral venue.	1 home; 0 neutral
RD_i, RD_j	Rest is the number of intervening Eastern-calendar dates since the previous game.	Varies; opener 3 days
TD_i, TD_j	Travel is the estimated distance between consecutive venue cities.	Varies; opener 0 km
TZ_i, TZ_j	Time-zone change compares dated venue offsets at the current tip-off.	Varies; opener 0 hours
F_i, G_i	The two empirical indices combine each side's rest, travel, and time-zone context.	Varies; positive

Symbol	Meaning	Value or range
y_i (y)	The observed outcome is one for a win and zero for a loss.	1 win; 0 loss
s_i (s)	The result sign equals twice the win indicator minus one.	+1 win; -1 loss
S_i (S)	Schedule strength combines the opponent's prior win rate and its prior opponents' win rates.	Varies; empty history 0.5
W_t	The smoothed prior win rate equals (prior wins + 1) divided by (prior games + 2).	Varies; no games 0.5
w_t, n_t	These are the wins and games observed for a team before the forecast cutoff.	Nonnegative integer counts
$\overline{W}_{\text{opp history}}$	This is the mean prior win rate of the opponent's previous opponents, counting repeats.	Varies; empty mean 0.5
LL_i, LL_j	Largest lead is the maximum points advantage achieved by each team during the completed game.	Nonnegative point counts
L_i (L)	Lead retention divides absolute final margin by the larger of both teams' largest leads.	Greater than 0; at most 1
Score	The box-score total is the weighted sum of the scaled box-score features.	Varies by team-game
Perf	The aggregate game performance factor (AGPF) applies a clipped sigmoid to the weighted Score.	From 0.000001 to 0.999999
PriorPerf	The baseline is the expanding mean of the team's prior game-performance values.	Varies; opener uses current Perf
P	The performance adjustment factor (PAF) compares current game performance with its prior baseline and clips the ratio.	From 0.25 to 4; opener 1
n	The prior-performance average divides the sum of a team's earlier game-performance scores by this count.	Number of prior team games
M_S	The schedule multiplier equals one plus the result sign times beta times schedule strength.	Varies by game
M_L	The lead multiplier equals one plus phi times the base-two logarithm of one plus lead retention.	Varies by game
M_P	The performance multiplier equals one plus the result sign times lambda times (P minus one).	Varies by game
Δr	The rating change equals the base outcome error multiplied by the three adjustment factors.	Varies by game

Box-score features and counts

Symbol	Meaning	Value or range
X_1	The feature measures point differential per 100 estimated team possessions.	Varies by team-game
X_2	The feature credits three-point makes with half an additional made shot.	Varies by team-game
X_3	The feature measures points per twice the estimated shooting opportunities.	Varies by team-game
X_4	The feature divides offensive rebounds by offensive rebounds plus opponent defensive rebounds.	Varies by team-game
X_5	The feature divides defensive rebounds by defensive rebounds plus opponent offensive rebounds.	Varies by team-game
X_6	The feature divides assists by player turnovers, using a denominator of one for zero turnovers.	Varies by team-game
X_7	The feature divides player turnovers by field-goal attempts plus 0.44 free-throw attempts plus player turnovers.	Varies by team-game
X_8	The feature divides steals by estimated opponent possessions.	Varies by team-game
x_k	The scaled features follow (X1/10, X2, X3, X4, X5, X6/3, -X7, X8).	Fixed transformation
$\mathrm{Poss}, \mathrm{Poss}_O$	Estimated possessions equal field-goal attempts plus 0.44 free-throw attempts minus offensive rebounds plus player turnovers.	Varies; coefficient 0.44
PTS, PTS_O	These counts record points scored by the team and its opponent.	Points in the completed game
FGM, FGA	These counts record field goals made and attempted.	Nonnegative integer counts
$3PM$	This count records made three-point field goals.	Nonnegative integer count
FTA	This count records free throws attempted.	Nonnegative integer count
$OREB, DREB$	These counts record offensive and defensive rebounds; an O subscript denotes opponent counts.	Nonnegative integer counts
AST, TOV, STL	These counts record assists, summed player turnovers, and steals.	Nonnegative integer counts

Fitting, evaluation, and travel notation

Symbol	Meaning	Value or range
θ	The fitted parameter vector contains seven scalar coefficients and seven independent feature-weight logits.	14 free coordinates
$\mathcal{L}(\theta)$	Training loss is the mean binary log loss across eligible training games.	0.6159695469 at fitted solution
N	The number of eligible games normalizes a stated fitting or evaluation score.	5370 release fit; 3690 new evaluation
u_1	This softmax logit parameterizes feature weight alpha1 relative to alpha8.	-2.435347556
u_2	This softmax logit parameterizes feature weight alpha2 relative to alpha8.	-2.251749454
u_3	This softmax logit parameterizes feature weight alpha3 relative to alpha8.	0.1515221307
u_4	This softmax logit parameterizes feature weight alpha4 relative to alpha8.	-2.349796664
u_5	This softmax logit parameterizes feature weight alpha5 relative to alpha8.	3.050615273
u_6	This softmax logit parameterizes feature weight alpha6 relative to alpha8.	4.657726628
u_7	This softmax logit parameterizes feature weight alpha7 relative to alpha8.	0.6873250356
u_8	The final softmax logit is fixed to identify the normalized feature weights.	0 (fixed)
$\varphi_1, \varphi_2, \ell_1, \ell_2$	These are the latitudes and longitudes of consecutive venue cities, expressed in radians.	Varies by venue
a_v	The Haversine intermediate combines the angular latitude and longitude differences.	Between 0 and 1
$v, o(v,t)$	The venue v and its UTC-offset function define a location's dated time-zone offset.	Venue dependent; hours
$\ln, \log_2, \exp$	These functions are the natural logarithm, base-two logarithm, and natural exponential.	Function definitions
clip	Clipping limits an input to its stated lower and upper bounds.	Bounds stated above

Carryover and probability corrections

Symbol	Meaning	Value or range
ρ	The season-start rating retains this fraction of the prior final deviation from 1500.	0.5
q_i, q_j	These are the default published injury-adjusted probabilities; their sum is one.	Between 0 and 1
σ, logit	The sigmoid and log odds convert between probability and an unconstrained score.	Defined in the text
$\mathbf{d}_i$	The standardized injury-head vector contains base log odds and prior player/report features.	19 coordinates
$b, \mathbf{w}$	The intercept and standardized-feature weights define the fitted logistic correction.	b = 0.00231409; weights in JSON
$\mathbf{w}_*, \eta$	The penalized vector includes the intercept; eta controls its squared-norm penalty.	20 coefficients; eta 0.0001
$\bar{m}$	Expected minutes average a player's earlier played observations with exponential decay.	Half-life 10 observations
a_i, a_j	Positive paired-comparison strengths are powers of ten of the rating divided by its scale.	$10^{r_i/c}, 10^{r_j/c}$
$A_{t,s}, B_{t,s}$	Team t's status s totals summarize expected minutes and expected minutes times player value.	Vary by report
u_k, μ_k, s_k, d_k	A raw injury-head input, its training mean and scale, and its standardized value determine one correction term.	$d_k = (u_k - \mu_k)/s_k$
PlayerScore	The descriptive player box-score total combines scoring, shooting, rebounds, assists, steals, blocks, fouls and turnovers.	Formula in the text
M, V	Discounted sums retain earlier player minutes and PlayerScore.	Half-life 10 observations
v_p	Prior value per minute is shrunk toward a fixed descriptive reference.	80 minutes at value 0.3
FTM, BLK, PF	These counts are free throws made, blocks and personal fouls.	Completed player box only
$q_{B,i}, \mathbf{f}_i$	The challenger uses this win probability and vector of prior team/injury covariates.	50 input coordinates
a_0, h_t, T	The boosted score adds numerical tree-leaf contributions to an initial log-odds score.	a0 = 0.21349012; 100 trees
ω	The blend weight controls how much the tree probability replaces base Elo.	1 in the current release
$s_{a,1}, s_{a,2}$	These are a player's minute shares in the last two observed rotations.	Fractions of team minutes

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